University College of Engineering

End Semester Examination

BCS303

Subject: Discrete Structures & Theory of Logic **Max Marks:** 80

Code: BCS303 **Duration:** 3 Hours

Instructions:

1. Attempt all questions.
2. Each section has specific instructions regarding the number of questions to be attempted.
3. Assume suitable data if necessary.

Section A — Section A

Short Answer Questions

10 questions x 2 marks each | Attempt all questions

Q1. What is a subset? **[2 Marks]**

K1 CO1

Q2. What is a Hasse Diagram in the context of Partial Orders and Ordered Sets? **[2 Marks]**

K2 CO1

Q3. A bijective function is a one-to-one correspondence between a set of distinct elements, where each element in the domain maps to exactly one element in the range, and every element in the range is mapped to by exactly one element in the domain. **[2 Marks]**

K1 CO2

Q4. Boolean Algebra axioms: **[2 Marks]**

K1 CO2

Q5. What is a logical fallacy? **[2 Marks]**

K1 CO3

Q6. What quantifier precedes the noun in a statement is the type of quantification used? **[2 Marks]**

K2 CO3

Q7. What is a subset that is closed under the group operation? **[2 Marks]**

K1 CO4

Q8. What is Homomorphism **[2 Marks]**

K2 CO4

Q9. What is a bipartite graph? **[2 Marks]**

K1 CO5

Q10. What is the Pigeonhole Principle? **[2 Marks]**

K1 CO5

Section B — Section B

Long Answer Questions

5 questions x 5 marks each | Attempt all questions

Q11. **[5 Marks]**

OR

a. {1, 2, 3, 5, 6, 10, 15, 30} **[5 Marks]**

b. What is a Distributive Lattice? **[5 Marks]**

Q12. **[5 Marks]**

OR

a. Create a Karnaugh Map with columns representing the variables x, y, z, w and rows representing the output values. **[5 Marks]**

b. What is the growth rate of the function f(x) = x^2 and g(x) = 2^x as x approaches infinity? **[5 Marks]**

Q13. **[5 Marks]**

OR

a. What is the Theorem of Deduction for statement calculus and explain the use of Modus Tollens? **[5 Marks]**

b. P (S (H (X))) **[5 Marks]**

Q14. **[5 Marks]**

OR

a. What is the Lagrange's Theorem for finite groups? **[5 Marks]**

b. What is a cyclic Group? **[5 Marks]**

Q15. **[5 Marks]**

OR

a. What is the difference between Euler paths and Hamiltonian paths? **[5 Marks]**

b. What is Graph Isomorphism? **[5 Marks]**

Section C — Section C

Descriptive Questions

5 questions x 7 marks each | Attempt all questions

Q16. **[7 Marks]**

OR

a. What are Lattices? **[7 Marks]**

b. What is the relation R defined as R(x,y) = {z | x ∧ y}, where z is a number? **[7 Marks]**

Q17. **[7 Marks]**

OR

a. What type of function is the one that is inversely proportional to another function? **[7 Marks]**

b. Theorems of Boolean Algebra: **[7 Marks]**

Q18. **[7 Marks]**

OR

a. What is the relationship between Satisfiability and Contradiction in Boolean logic? **[7 Marks]**

b. What is the Inference Theory of Predicate Logic? **[7 Marks]**

Q19. **[7 Marks]**

OR

a. What are Rings and Fields? **[7 Marks]**

b. Subgroups and Cosets **[7 Marks]**

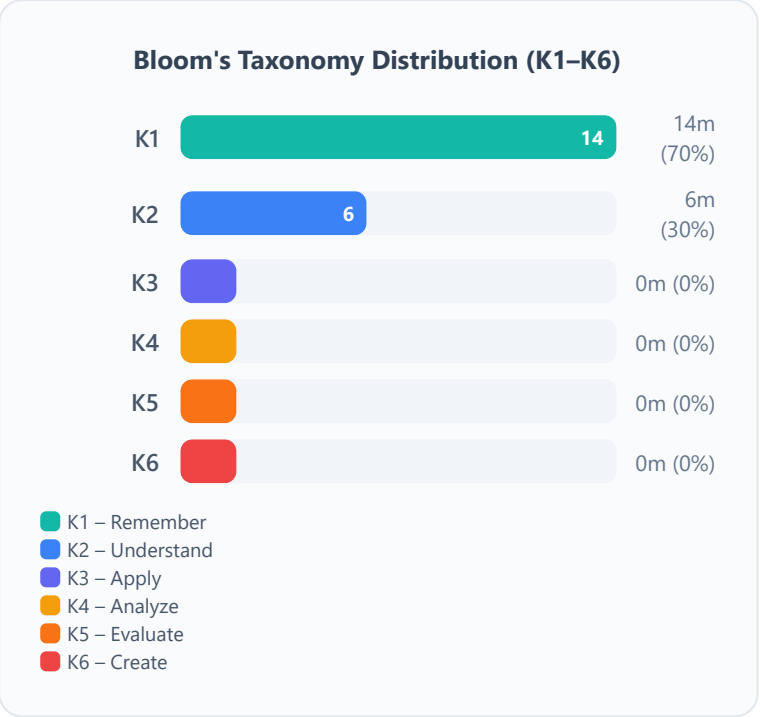
Q20. **[7 Marks]**

OR

a. What is a planar graph? **[7 Marks]**

b. Explain various counting techniques in Combinatorics. Illustrate the application of the Pigeonhole Principle in solving discrete mathematical problems. **[7 Marks]**

Question Paper Analysis



Bloom's Taxonomy Summary		
Level	Marks	%
K1	14	70%
K2	6	30%
K3	0	0%
K4	0	0%
K5	0	0%
K6	0	0%

Course Outcome Summary		
CO	Marks	%
CO1	4	20%
CO2	4	20%
CO3	4	20%
CO4	4	20%
CO5	4	20%